EEG-Based Emotion Recognition Using SVM

Ram Avtar Jaswal and Sunil Dhingra

Abstract Emotional recognition and detection has become one of the fastest emergent areas which has encouraged the researchers to do more research in this area. In this paper, we described an EEG-based emotion recognition method which identifies the human emotional states. The proposed method used the DEAP database and developed a supervised machine learning algorithms to recognize the emotional from EEG signal. The Welch's transformer is used to preprocess the EEG signal to obtain the alpha, delta, theta, beta waves and gamma waves. Others attributes such as mean, variance, standard deviation, kurtosis and skewness are also collected. After using SVM and XGbooster supervised machine learning algorithms, appropriate emotional state has been detected. The proposed method achieved the 96.2% response rate for happy, 94.8% for neutral, 93% for sad and angry. The proposed method has been compared with existing method.

Keywords Emotions · Electroencephalography · Deep convolutional neural network

1 Introduction

Emotions are the basis of human daily life and play a significant part in human intellect such as in coherent decision making, human intelligence, understanding and interpersonal communication. The emotion state of a person can be clearly recognized on the basis of the audio/visual signals (external expressions), physiological signals (internal expressions) and person perception. Human–computer interaction (HCI) systems are getting attention of researchers to recognition the emotions of

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human being [1]. Recorded physiological signals of participants play an important role in this area, and after analyzing these signals, emotions of human can be recognized. Physiological signals have been recorded from central and autonomic nervous system. In central nervous system, signal from spinal cord and brains has been recorded, whereas in autonomic system physiological signal also contains other regulating physiological functions such as heart rate and arousal. There are different methods by which physiological signals have been recorded. The common signals that measured the emotions are galvanic skin response (GSR), electromyography (EMG), hear rate (HR), respiration rate (RR) and electroencephalography (EEG). In GSR, the level of emotional increases with respect to the level of arousal, whereas in EMG, it is negatively associated the valence emotions with frequency of muscle tension. In case of heart rate, the rate increases with negative emotions like fear, whereas RR changes irregularly with emotions like anger. Electroencephalography (EEG) is a one of techniques which is used the neuro-imaging that measure potential voltage changes through electrical impulses in triggering neurons, detect them in the scalp and record them with a device with electrode samples [2–4]. EEG signals are classified according to frequency and can be separated in five frequency bands: delta band (0.5–3 Hz), theta band (4–7 Hz), alpha band (8–13 Hz), beta band (14–30 Hz) and gamma band (above 30 Hz). These frequency band can be related with specific types of activities such as finger movements, falling asleep, active thinking and problem solving. EEG, which used in brain–computer interface (BCI), facilitates communication between computer and human being independently [5]. To understand the human state of mind, application of affective computing becoming increasingly popular day by day.

2 Related Work

The method of expressing emotions is the bipolar model proposed by Russell in [6] with arousal and valence dimensions. The value range of valence extends from −ve to +ve and range of aroused from "not-aroused" to "excited". This 2D model can recognize emotional labels and even also define emotions without isolated emotional label. However, the feelings of fear and anger had not varied in model as both emotional have the same −ve valence level and high-arousal values.

Mehrabian and Russell in [7] proposed 3D Pleasure (Valence)-Arousal-Dominance (PAD) model. The model's "pleasure-displeasure" level corresponds to the value of 2D model which measure the level of pleasure-emotional. The range of "Arousal Non-arousal" corresponds to the value of the Arousal and "Dominance Submissiveness" correspondence to the dimension of emotional control. The dominant dimension finds more sensitive labels in 3D model. As surprise and happiness have high arousal and positive valence value, still model is able to differentiate in these emotional using the degree of dominance.

Emotional cognition has received a lot of attention over the past decade as it is directly related to physiology, psychology, marketing and therapy. Previous studies in [8–13] proposed different types of frequency domains, statistical properties, time domains and various feature extraction methods. In [14], author proposed a method combine six statistical properties in time domain of EEG signals to identify human emotions. The proposed method removed the noise and repetitive channels using the Relief-F and PCA algorithms. The method applied the SVM training to DEAP data and identified emotions. The resultant accuracy was 81.87%.

In the [15], authors tried to diminish the size of the aspect vectors to improve classification accuracy. The proposed method used primary component analysis (PCA) to rule out symptoms. PCA and SVM method applied to SEED dataset which classify positive, negative, and neutral emotions, and resultant accuracy was 85.85%.

In [16], authors experimented different kind of classifications on various functions. The authors collected statistical properties, entropy, energy density, frequency properties and wavelet energy. Researchers selected 14 electrodes from the DEAP database and collected 168 properties for every subject. Authors applied the decision tree, SVM, KNN classifiers, and accuracy result was 77.54% for valence, 79% for dominance and 78.88% for arousal.

The authors in [17] discussed multi-classification as a way to improve accuracy and had an excellent collaboration. First, the researchers collected several properties of the time–frequency field and the time–frequency properties of EEG signals. The researchers use PSO algorithm for feature selection. Researchers achieved the 76.67% accuracy rate.

Author proposed the DGCCA-AM method in [18] which provides a combination of the recognition characteristics of emotions into five categories. Authors used the short-time Fourier transform (STFT) and non-overlap of Hanning windows for 4 s to collect the differential entropy (DE) characteristics of the EEG signals. The proposed method is analyzed using a public dataset SEED-V. The experimental recorded an accuracy of 82.11%.

Authors in [19] projected a method based on the asymmetry index (AsI) to estimate the level of emotion in streaming music videos. AsI estimated the level of emotional by measuring the amount of mutual information in two frontal lobes. The low emotional signal segments are filtered out after calculating the ASI and then wavelet translation is applied. After extracting the features, SVM classification has been applied and obtained accuracy rate of 70.5%.

Authors in [20] projected decision tree and a rule-based classification algorithms which used EEG signals to identify emotions in four categories, i.e., happiness, sadness anger and relaxation. The proposed method collected the characteristics of the frequency and time domain of DEAP dataset. Then rules classification model generates 10 emotion classification rules and achieves an average precision of 81.64% for the relaxed emotion class.

Many studies have done extensive research on the recognition of emotions and have shown that the classification of emotions is more precise. In [21], authors used deep learning to categorize four emotions and applied functions to obtain the time, entropy and frequency domain. The precision of the time domain attributes was 93.48%, the precision of the frequency domain attributes was 92.44% and the precision of the entropy-based attributes was 93.75%.

Authors in [22] classified the emotions into two levels of arousal, valence. The proposed method involves the analysis of non-stationary nonlinear time series of EEG signals. The authors applied artificial neural network (ANN) classification on DEAP dataset. The results of arousal accuracy are 75% and valence accuracy of 72.87%.

In [23], authors applied STFT transform to obtain the high-order crossing (HOC), HHS spectrum of the delta, alpha, beta and gamma bands from DEAP dataset. Then authors used the mRMR algorithm to remove non-informative features. The results show that the overall accuracy is less than 60% even mRMR improves the accuracy of the results.

3 Proposed Work

The proposed method consists of data acquisition, preprocessing, feature extraction and classification step. The proposed method used EEG signals the DEAP dataset [24]. The preprocessed DEAP dataset are analyzed using various functions of time and frequency domains. The proposed method develops a supervised machine learning model to divide emotional states into four categories. Figure 1 illustrations the proposed framework to recognize the emotions.

3.1 Data Acquisition

The first step of proposed method is data acquisition. Depending on the capacity of the system, different sampling rates can be used to obtain EEG signals in system. In the proposed method, DEEP dataset has been used to analyze the emotion through EEG signals. Brain signals of 32 subjects were retrieved from EEG device at 500 Hz sampling rate in the DEAP database. The dataset has two type of physiological signal data: preprocessed and raw data. In raw data, different results may be obtained by using different preprocessing function. To maintain consistency, the proposed

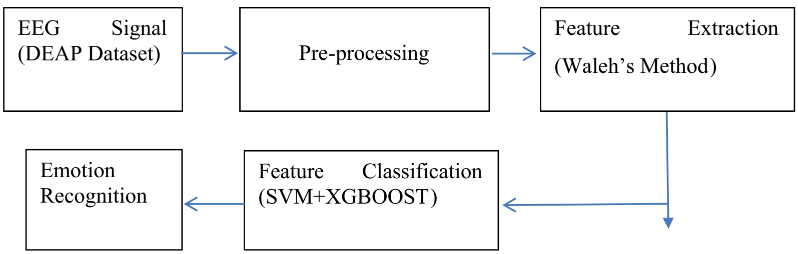

Fig. 1 Proposed method framework

method uses the Python preprocessed data. Figure 1 shows the placement of the EEG electrodes in dataset. The preprocessed dataset consists of 32 EEG signals (128 Hz) and 8 peripheral signals. There are two type of array contain in data: first for data and other for the emotional label. The data is applied to the classification algorithm independently of the preprocessing step. The method used the Python's pickle library to read EEG data. Using the Python's numpy library, data can be encoded in the different files which can be used to perform various binary classifications to distinct categories of valence, arousal, dominance and liking emotions.

3.2 Preprocessing and Feature Extraction

The values of arousal and valence are divided into two classes: high arousal (HA) and low arousal (LA). The combinations of valence and arousal can be converted to emotional states: high-arousal positive valence (Excited, Happy), low-arousal positive valence (Calm, Relaxed), high-arousal negative valence (Angry, Nervous) and low-arousal negative valence (Sad, Bored). These two-dimensional emotional space can be divided into four areas: HAHV, HALA, LAHV and LVLV. Table 1 gives the mean and standard deviation value of HAHV, LAHV, HALV and LALV. The scatter plot of the HAHV, LAHV, HALV and LALV is shown in Fig. 2.

Table 1 Mean, standard deviations (STD) value

	Valence	Arousal
HAHV	Mean 7.12 STD 1.04	Mean 6.81 STD 0.84
LAHV	Mean 6.67 STD 1.09	Mean 3.87 STD 1.09
HALV	Mean 3.19 STD 1.23	Mean 6.79 STD 0.96
LALV	Mean 3.57 STD 1.12	Mean 3.47 STD 1.19

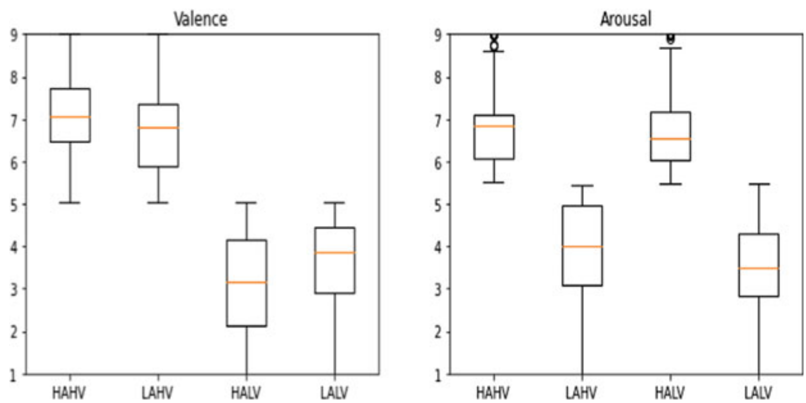

Fig. 2 Values of HAHV, LAHV, HALV, LALV

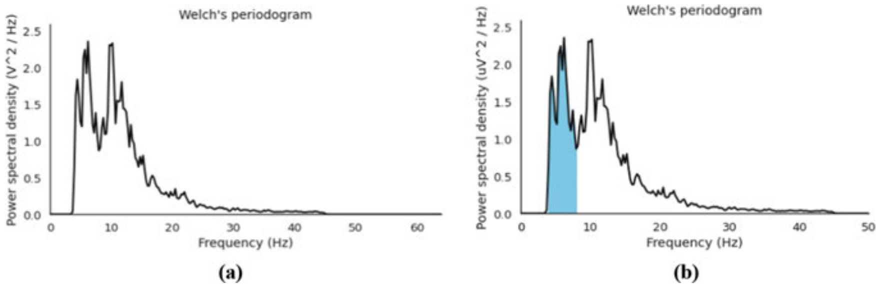

Fig. 3 Power spectral density **a** frequency band, **b** theta band

The frequency band of EEG signal is divided into four bands: theta (θ: 4–7 Hz), alpha (α: 8–13 Hz), beta (β: 14–30 Hz) and gamma (γ: 31–50 Hz). The Hanning window is applied to each EEG channel and power spectral density (PSD) calculated by Welch's method which is shown from Figs. 3, 4, 5, 6, 7, 8 and 9.

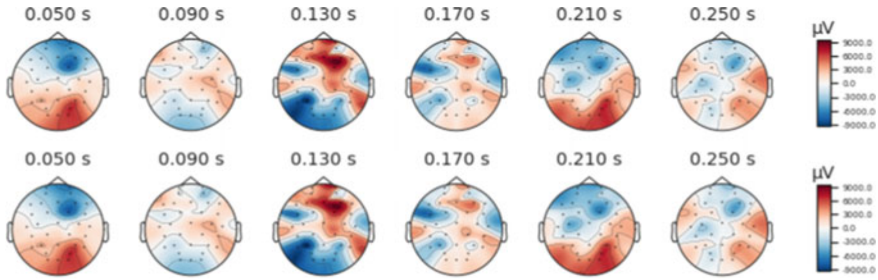

Fig. 4 Theta band

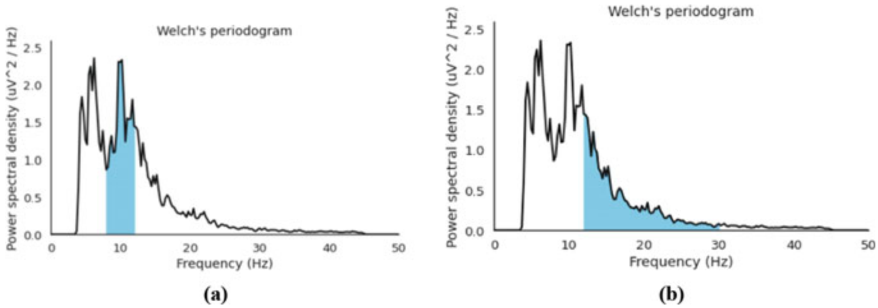

Fig. 5 Power spectral density **a** alpha band, **b** beta band

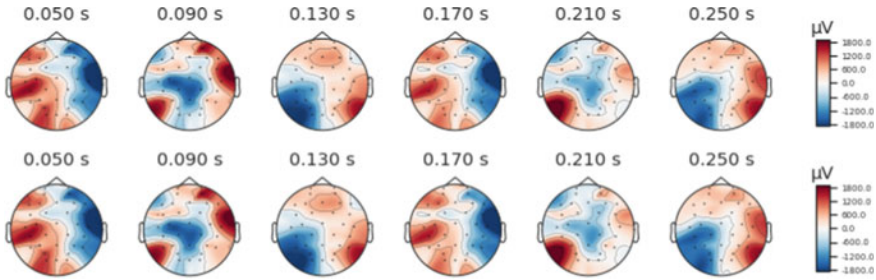

Fig. 6 Alpha band

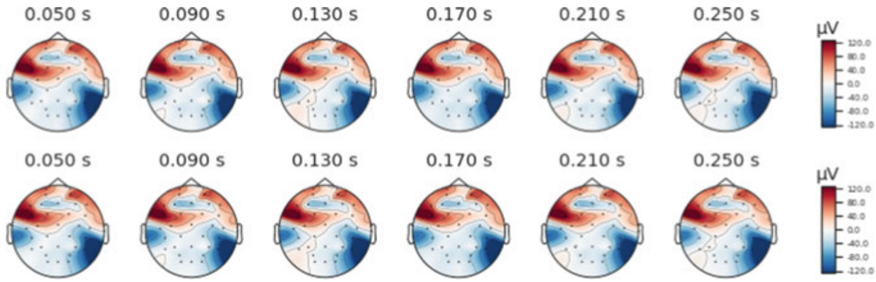

Fig. 7 Beta band

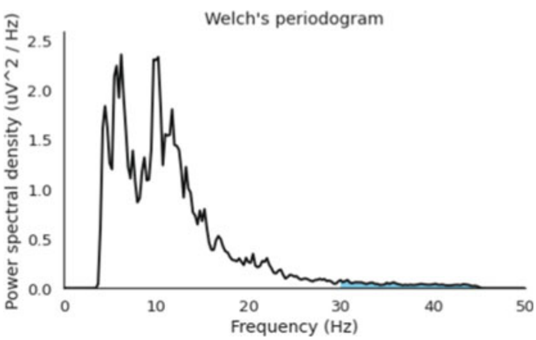

Fig. 8 Power spectral density-gamma band

3.3 Emotions Classification

The classification first uses sample emotion data for training and then classifier recognizes human emotions after training. For classification, SVM with XGBooster architecture has been used in the proposed method.

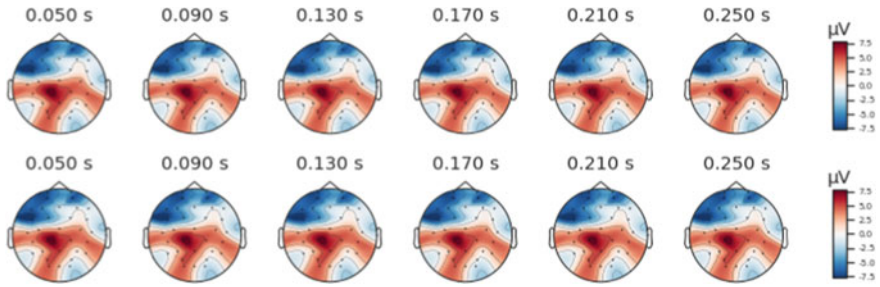

Fig. 9 Gamma band

Table 2 Confusion matrix

Predictable emotion	Experimental emotion				Accuracy (%)
	Happy	Neutral	Sad	Angry	
Happy	96.2	0.24	1.1	1.1	96.2
Neutral	0.3	94.8	1.2	2.2	94.8
Sad	2.4	2.46	93.4	3.4	93.4
Angry	1.2	2.5	4.3	93.3	93.3

4 Results and Discussion

The confusion matrix technique is used to check the performance of classification algorithms. If the number of observations for each class is not equal, or the dataset contains multiple classes, the accuracy of the classification will only be misleading. The confusion matrix calculation can provide a good understanding of the types of errors generated by the classification model. This matrix can be used for problems that can be understood by two classes, but it can also be easily applied to problems with three or more class values by adding rows and columns to a messy matrix. Table 2 gives the values of the confusion matrix used to test the accuracy of the data used. For example, if user is sad, the correct percentage is 93.4%.

5 Comparison with Existing Methods

Finally, Table 3 compares the proposed work with other work using the same DEAP dataset. The proposed work gives better results with an accuracy rate of 94.425%. However, Ahirwal and Kose classification gives better results than Nawaz et al. and Li et al. However, due to the number of levels and calculations, method can be time-consuming. The results given by SVM are better than to ANN.

Table 3 Comparison of proposed method

Authors	Algorithm classification	Accuracy (%)
Nawaz et al. [16]	SVM, KNN, DT	77
Ahirwal and Kose [21]	ANN	93
Li et al. [17]	SVM	76.67
Proposed method	SVM	94.425

6 Conclusion and Future Scope

In this paper, we have proposed an emotion recognition system-based EEG data. The proposed method performed well in terms of accuracy. It can be seen from the results that the happy state achieves 96.2% results compared to other states. In a neutral state, 94.8% recognition rate was observed. However, detection rate of angry and sadness has been reduced by 1.1 and 1.2%, respectively. In the future, we will consider using more physical parameters for future development and improvement of the system which can increase the recognition rate of the system. Currently only four basic emotions are calculated, more emotions, i.e., fear, stress, etc., may be calculated in the future.

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